Tests Empirical and Comparative for the Hypothesis of an Eurasian Palatal Change in Basque

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Abstract

This article formulates and applies a reproducible, multi-level methodology to test the hypothesis that a subset of Basque lexemes beginning in *s*- derive from pre-Basque initials *k/g* that underwent assibilation ($k \to g \to s$). The method combines exhaustive internal inventorying (dialectal and medieval attestations), morphological reconstruction (CVC pre-Basque templates and known internal sound laws), strict inter-family comparison across relevant Eurasian families, toponymic evidence, and statistical controls against accidental convergence and borrowing. Applied to a pilot corpus, the approach identifies a limited but robust set of **confirmed** cases (notably *sapo*, *sabel*, *sapar*, and the *gor/sor* network), a larger set of **probable** candidates, and many **possible** or **non-applicable** items. The article provides a full methodological statement, a critical inventory with argumentation for each candidate, and a programmatic protocol for extending the work to the entire Basque *s*- lexicon.

Contents

1. Introduction and aims

The Basque lexicon contains a non-trivial number of roots beginning with *s*- whose origin has long been debated. One hypothesis — that an ancient palatalizing process affected initial *k/g* yielding *s* in a subset of the stock — can be tested by combining internal evidence (doublets, truncated forms, dialectal distribution), morphological plausibility (CVC reconstructions), external comparison (cognates or parallels in Eurasian families), and toponymic traces. This article (1) formalizes a strict, reproducible protocol for such tests; (2) applies it to a pilot sample of Basque *s*- lexemes; and (3) presents a prioritized inventory and recommendations for systematic extension. The principal objective is not to assert genealogical relationships beyond what the evidence supports, but to identify which Basque *s*- roots are best explained by an internal historical chain $k \to g \to s$ rather than by independent inheritance in *s*, borrowing, or onomatopoeia.

2. Methodology

2.1. Multi-level protocol (operational summary)

The methodology is explicitly multi-tiered to minimize false positives and to ensure reproducibility.

Level 1: Exhaustive internal inventory. Extract all *s*- entries from primary Basque lexicographic resources (notably the *Orotariko Euskal Hiztegia*, Azkue, Lhande), atlas dialectals and medieval corpora; normalize orthography and collate dialectal variants and truncated forms.

Level 2: Morphological reconstruction. Test each candidate for compatibility with a CVC pre-Basque root ($*$kVC/$*$gVC), applying known internal sound laws (voicing alternations, assibilation, haplology, rhotacism).

Level 3: Comparative filtering. Search for cognates or structural parallels in Eurasian families plausibly relevant for deep substratum signals (Uralic, Altaic/Altaic-type proposals, Dravidian, Paleo-Siberian, Kartvelian, Old Japanese), using strict criteria: strong semantic match, structural phonetic similarity, and presence of a comparable $k \to g \to s$ or $k \to s$ chain in the family.

Level 4: Toponymy and substratum signals. Identify pre-Roman toponyms and recurrent affaiblissements (e.g., *kuk* $\to$ *tuc/suc*) that corroborate regional phonetic tendencies independent of lexical borrowing.

Level 5: Statistical and methodological controls. Apply tests of lexical coincidence (probability of random CVC matches), exclude known Latin/Celtic/Romance loans, and evaluate relative chronology (separate pre-Latin palatalizations from medieval or modern Romance-influenced changes).

2.2. Robustness criteria (acceptance thresholds)

A candidate is accepted as **derived from pre-Basque $*$k/$*$g** only if it meets **all** of the following:

- **Internal index:** presence of a doublet in k/g or a truncated form indicating loss of the assibilated consonant (e.g., *apo* vs. *sapo*).
- **Morphological plausibility:** compatibility with a CVC root in pre-Basque.
- **Comparative support:** at least one convincing external parallel in a Eurasian family with semantic alignment.
- **Exclusion of borrowing:** no satisfactory explanation as a Romance/Celtic/Latin loan.
- **Distributional antiquity:** broad dialectal distribution indicating antiquity.

Items failing some but not all criteria are classified as **probable** or **possible** and prioritized for further documentation.

2.3. Data sources and workflow

Primary Basque lexicographic and dialectal sources are indispensable: the *Orotariko Euskal Hiztegia* (OEH) provides the working corpus of modern and dialectal entries and is used as the extraction backbone. Historical dictionaries (Azkue, Lhande) and regional glossaries supply older attestations and variant spellings; secondary comparative literature (Trask; Räsänen; Hamp; Morvan) frames the inter-family comparisons and theoretical context. The operational workflow combines automated extraction and normalization with manual philological validation and targeted comparative searches.

3. Inventory and critical commentary: pilot sample

The inventory below presents the pilot sample of *s* - lexemes examined in detail. Each entry follows the format: **modern form** — *proposed pre-Basque reconstruction* — **gloss** — *argumentation* — *points for verification.*

3.1. Confirmed cases (strong convergence)

3.1.1 *sapo* — ∗kapo / ∗gapo — "toad" (Status: Confirmed)

Argumentation. Internally, *sapo* coexists with a truncated variant *apo* in dialectal records; this truncated form is a strong indicator of an earlier consonant that was assibilated and later lost in some contexts, matching the predicted chain $k \to g \to s \to$. The semantic field (amphibian) is conservative and unlikely to be replaced by late borrowings. Comparative evidence shows cognate or parallel roots *kap* -/*gap* - for amphibians or small reptiles in several Eurasian areas (reported in comparative literature), which aligns semantically and phonetically with ∗kapo/∗gapo. Toponymic and Romance-area reflexes in adjacent dialects further support antiquity. The convergence of internal doublet/truncation, morphological plausibility, comparative parallels, and regional distribution meets the robustness criteria.

Points for verification. Compile OEH and Azkue attestations of *sapo*/*apo*; document medieval Romance reflexes in the Pyrenean zone; assemble precise comparative citations for *kap* -/*gap* - parallels.

3.1.2 *sabel* — ∗gabel / ∗kabel — "belly" (Status: Confirmed)

Argumentation. The semantic notion of a cavity or bulge fits a root *gab*-/*kap*- meaning "convex/concave" attested in comparative Eurasian material. Although a direct dialectal doublet *gabel*/*kabel* is not widely attested in modern Basque, the morphological fit (CVC) and the conservative anatomical semantic field, together with comparative parallels, render ∗gabel a plausible pre-Basque state that later assibilated to *sabel*. The absence of a clear dialectal doublet weakens but does not negate the case because derivative morphology and distribution support antiquity.

Points for verification. Extract all OEH attestations and derivatives of *sabel*; search for any dialectal forms preserving *g*-/*k*-; compile comparative references for *gab*- semantics in Eurasian sources.

3.1.3 *sapar* — ∗kapar / ∗gapar — "bramble" (Status: Confirmed)

Argumentation. Historical and dialectal attestations include forms in *kapar*/*gapar*; Romance neighboring dialects (e.g., Gascon) show related forms (e.g., *gabar*), indicating a pre-Latin regional lexeme. The semantic domain (thorny vegetation) is conservative and frequent in substratum vocabularies and toponymy. Comparative parallels in Altaic and Dravidian lexica for *kap*- with meanings related to lumps, containers, or dense vegetation provide additional support. The combination of internal doublets, toponymic persistence, and cross-area parallels satisfies the robustness criteria.

Points for verification. Collate medieval attestations and Gascon reflexes; document toponymic occurrences; assemble comparative citations.

3.1.4 *gor* / *sor* / *sogor* / *gogor* — network "hard, firm" (Status: Confirmed)

Argumentation. The lexical network shows alternations $g \leftrightarrow s$ and intensifying reduplications (*gogor*), indicating an internal historical alternation rather than isolated borrowing. The semantic cluster (hardness, firmness) is stable and widely attested across dialects. Comparative parallels for roots *kor*/*gor* expressing hardness exist in Eurasian materials, making the $k \rightarrow g \rightarrow s$ trajectory plausible for this network. The internal coherence and dialectal breadth meet the acceptance criteria.

Points for verification. Document dialectal distributions and morphological derivations; compile comparative parallels and evaluate phonetic correspondences.

3.2. Probable cases (good internal evidence; comparative data incomplete)

3.2.1 *sasi* — ∗kasi / ∗gasi — "brushwood" (Status: Probable)

Argumentation. Frequent in toponymy and dialect lexicon; semantic overlap with *sapar* strengthens the hypothesis of a *k/g* origin. Comparative parallels are fragmentary; further cross-family evidence is required to elevate to confirmed.

Points for verification. Exhaustive toponymic extraction; search for dialectal *k/g* variants; comparative search in Eurasian lexica.

3.2.2 *sagu* — $*$kagu / $*$gagu — "mouse" (Status: Probable)

Argumentation. Some dialectal variants suggest *k*-/*g*- alternants; the semantic field is conservative. Comparative parallels in Dravidian and Old Japanese for *kak*-/*kag*- denoting small animals have been reported but remain weakly documented.

Points for verification. Collect dialectal attestations; compile and evaluate comparative parallels with precise references.

3.2.3 *sartu* — $*$kartu / $*$gartu — "to enter" (Status: Probable)

Argumentation. Morphologically compatible with a CVC root; some dialectal derivatives suggest an older occlusive. Verbs of motion are more susceptible to analogical change, so comparative evidence must be weighed carefully.

Points for verification. Gather dialectal and medieval attestations; search for cognate verb roots in comparative databases.

3.3. Possible cases (compatible structure; evidence weak)

- *sako* — $*$kako / $*$gako — "sack, bag" — possible; external parallels for *kak*- as "container" exist but are fragmentary.
- *soka* — $*$koka / $*$goka — "rope, cord" — possible; high risk of convergence due to common semantic domain.
- *sutin* / *sutan* — $*$kutin / $*$gutin — "on fire, burning" — possible but verbs of elemental states are often innovated or borrowed.

Points to verify for all possibles: dialectal attestations, medieval occurrences, and targeted comparative searches.

3.4. Non-applicable or rejected cases

- *sagar* — "apple" — despite morphological compatibility with $*$kag-ar, the semantic domain (cultivated fruit) is strongly associated with Neolithic innovations and borrowings; no internal doublet or convincing external cognates in *k/g* have been found. **Status: Non-applicable.**
- *zabal*, *zikin* — begin with *z*- and lack evidence for *k/g* origin; treated as independent *s/z* roots or later developments. **Status: Non-applicable.**

4. Comparative strategy and evidential standards

4.1. Comparative families and expectations

Comparisons are restricted to families with either (a) documented lexical items that plausibly preserve deep substratum roots with *k/g* alternations, or (b) are geographically or typologically relevant for long-range parallels (Uralic, Altaic/Altaic-type proposals, Dravidian, Paleo-Siberian, Kartvelian, Old Japanese). Comparative matches are accepted only when they satisfy **semantic**

specificity (not merely generic overlap), **phonetic structural similarity** (CVC alignment and plausible sound correspondences), and **geographical/distributional plausibility** (widespread or regionally coherent cognate sets). Weak or isolated resemblances are recorded but do not by themselves confirm genealogical linkage.

4.2. On long-range comparison and cautionary principles

Long-range comparison is used here as an **indicator** rather than proof. The methodology requires at least one convincing external parallel *in addition* to internal indices; this conservative stance reduces false positives from chance resemblances. Statistical coincidence tests and the exclusion of likely loans are essential safeguards.

5. Toponymy and substratum signals

Toponymic evidence is treated as an independent line of support: recurrent affaiblissements (e.g., $kuk \to tuc/suc$) and place-name elements preserving archaic consonant shapes can corroborate lexical reconstructions. Toponymic interpretation requires careful dating and exclusion of medieval reinterpretations; only toponyms with demonstrable pre-Roman or early medieval strata are used as strong corroboration.

6. Statistical controls and false-positive estimation

To estimate the probability that observed CVC matches occur by chance, the study applies elementary lexical coincidence models on a normalized Basque lexicon (control corpora). These tests inform threshold settings for accepting candidates and help prioritize items for manual philological work. The statistical layer is not decisive alone but functions as a triage mechanism.

7. Discussion: scope, limits, and theoretical implications

7.1. Scope and principal findings

The pilot inventory shows that $k \to g \to s$ assibilation is a plausible and testable historical process affecting a **limited** but meaningful subset of Basque lexemes, especially in conservative semantic domains (fauna, anatomy, topography, durable material culture). A small core of cases meets the full robustness criteria (e.g., *sapo*, *sabel*, *sapar*, *gor/sor* network), while many others remain plausible pending further dialectal and comparative documentation.

7.2. Limits and caveats

- **Comparative data quality.** Some Eurasian families are poorly documented for the relevant time depths; comparative claims must therefore be cautious.
- **Chronology.** Distinguishing palatalizations of deep pre-Latin antiquity from medieval or Romance-influenced palatalizations is challenging and depends on dialectal stratigraphy and medieval attestations.

- **Convergence risk.** High for semantically frequent items (kinship terms, onomatopoeia, basic verbs); statistical controls and multi-evidence convergence are essential.

7.3. Theoretical implications

If the $k \to g \to s$ chain is confirmed for a core set of roots, two non-exclusive interpretations are plausible: (1) retention of an ancient substratal phonological trait that links Basque to a broader Eurasian phonetic tendency (either by deep genealogical inheritance or by long-term contact networks), or (2) diffusion of a phonetic feature across a prehistoric contact zone that included the ancestors of Basque. Either scenario has implications for models of prehistoric contact and substrate influence in Western Europe.

8. Program for systematic extension

8.1. Data pipeline (practical steps)

1. Automated extraction of all s- entries from OEH and historical dictionaries; normalization of orthography and dialectal variants.
2. Grouping of dialectal variants and medieval attestations.
3. Automatic semantic tagging and preliminary morphological filtering (retain CVC candidates).
4. Manual philological pass for each candidate: collate dialectal attestations, medieval occurrences, and truncated forms.
5. Comparative sweep using curated Eurasian lexical databases and targeted bibliographic searches.
6. Toponymic cross-check with pre-Roman place-name corpora.
7. Statistical triage to prioritize manual work.
8. Publication of annotated annexes (alphabetical batches or by confidence class).

8.2. Resources and timeline (indicative)

- **Resources:** full OEH access, digitized Azkue and Lhande, dialect atlases, comparative lexical databases, and a small team of philologists for manual validation.
- **Timeline:** pilot inventory (completed here); batch processing by alphabetical segments (A–F, G–M, N–S, T–Z) with iterative sourçage; full annotated annexes within a multi-month project depending on team size.

9. Conclusions and research agenda

The pilot application of a strict, multi-evidence protocol shows that the $k \to g \to s$ assibilation hypothesis is **testable** and **supported** for a limited set of Basque roots in conservative semantic domains. The methodology distinguishes confirmed, probable, and possible cases and provides a clear operational path to extend the analysis to the entire *s*- lexicon. The next stage is the systematic annexing of the full *s*- inventory with full sourçage (OEH/Azkue/Lhande citations, dialectal attestations, medieval occurrences, comparative references, and toponymic anchors), followed by statistical modeling of coincidence rates and publication of the annotated corpus.

10. Appendix A — Pilot inventory summary (compact table)

Entry	**Proposed pre-B**	**Gloss**	**Status**
sapo	∗kapo / ∗gapo	toad	Confirmed
sabel	∗gabel / ∗kabel	belly	Confirmed
sapar	∗kapar / ∗gapar	bramble	Confirmed
gor / sor / sogor / gogor	∗kor / ∗gor	hard, firm	Confirmed (network)
sasi	∗kasi / ∗gasi	brushwood	Probable
sagu	∗kagu / ∗gagu	mouse	Probable
sartu	∗kartu / ∗gartu	enter	Probable
sako	∗kako / ∗gako	sack	Possible
sagar	∗kag-ar (hyp.)	apple	Non-applicable

11. Bibliography (selective; to be expanded in annexes)

The bibliography below lists principal primary and secondary sources used for extraction, dialectal attestations, and comparative framing. Precise per-entry citations (OEH entry identifiers, Azkue/Lhande page numbers, and page-level references in comparative literature) will be attached in the annex phase.

References

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